MIL-T-45603 (Ord)
31 December 1958
SUPERSEDING
FCS-862A
31 March 1958

MILITARY SPECIFICATION

TIMER, INTERVAL TD-114/M

1. SCOPE

1.1 This specification covers one type of interval timer designated, Timer, Interval TD-114/M, hereinafter referred to as "timer".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant

Military

MIL-P-116	Preservation, Methods of
MIL-B-131	Barrier Material; Water Vaporproof, Flexible
MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-C-7769	Cushioning Material, Bound
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electric Component Parts
MS 20003	Indicator-Humidity, Card, Three Spot Impregnated Areas

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DRAWINGS

Ordnance Corps

D7614285	Timer, Interval TD-114/M Schematic Diagram
E7620519	Timer, Interval TD-114/M

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED FREIGHT CLASSIFICATION RATINGS, RULES AND REGULATIONS

(Application for copies should be addressed to the Consolidated Freight Classification Committee, 202 Union Station, Chicago 6, Illinois).

NATIONAL MOTOR FREIGHT CLASSIFICATION No. A-2 AND No. 13

(Application for copies should be addressed to the American Trucking Association, 1424 Sixteenth Street, N.W., Washington 6, D.C.).

3. REQUIREMENTS

3.1 Material.- Material shall be in accordance with the drawings and specifications forming a part of this specification.

3.2 Electrical.

3.2.1 Continuity.- The timer shall demonstrate continuity in accordance with the electrical schematic drawing D7614285.

3.2.2 Dielectric strength.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the timer are subjected to an AC root mean square (r.m.s.) potential of 1000 volts plus twice the rated voltage at a frequency of 60 cycles per second (c.p.s.) for a period of not more than 30 seconds nor less than 5 seconds. Insulation resistance requirement shall be checked immediately after this requirement.

3.2.3 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (d.c.) applied.

3.3 Performance.- The timer, Drawing E7620519 when set-up in accordance with pretest information (section 4) shall meet the following requirements. (All symbols referred to in the following paragraphs are those shown on Drawing D7614285.)

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3.3.1 Operate time.

(a) Relay K1 shall be energized 14±4 seconds after the timer has been put in the operate condition and R11 set fully counterclockwise.

(b) Relay K1 shall be energized 33±7 seconds after the timer has been put in the operate condition and R11 set fully clockwise.

3.3.2 Recycle time.- When R11 is set that relay K1 energizes 22 ± 2 seconds after the timer has been put in the operate conditions, the timer shall exhibit such interruption and reoperation times as specified:

Interrupt time (Seconds)	Re-operate time (Seconds)
0.5	0-6
1.0	2-8
5.0	6-14
10.0	8-20

3.4 Environmental.

3.4.1 Vibration.- The timer shall suffer no damage and shall meet the performance requirements of this specification after vibrations as follows:

Vibration Frequency* (cycles per sec)	Total Excursion (inches)	Acceleration (g's)	Time (hrs ea direction)
2-10	0.060	---	1
11-20	0.030	---	2
21-400	---	2.5	2

*Uniform rate of change shall not be greater than 1 cycle per 3 seconds nor less than 1 cycle per 5 seconds.

3.5 Workmanship.- Workmanship shall be in accordance with the applicable workmanship requirements of Specification MIL-F-13926.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions

4.1.1 Contractor inspection.- Unless otherwise specified herein, the inspection required by this specification will be performed by the contractor.

4.1.2 Contractor quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. Results of all examinations and tests performed under the quality assurance system shall be made available to the Government. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the specification and other documents referenced in this specification. The contractor will

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not be restricted to the inspection station or to the method of inspection listed provided that the substitue procedures will provide, as a minimum the quality assurance required in the contractual documents. In cases of dispute as to whether certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced therein.

4.1.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures of poor practices which might have an affect upon the quality of the product will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Inspection provisions.- Applicable quality assurance provisions common to all fire control instruments as specified in Specification MIL-F-13926 shall apply.

4.2.1 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.2.2 Examination.- The classification of defects in Tables I, II and III shall constitute the minimum inspection requirements to be performed prior to acceptance.

4.2.2.1 All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS
(Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105)

Major AQL 1.0	Req. Par.	Q.A. Par.	Test Method
1. Continuity	3.2.1	4.3.1	SME
2. Dielectric strength	3.2.2	4.3.2	SME
3. Insulation resistance	3.2.3	4.3.3	SME
4. Operate time	3.3.1	4.3.4.2	SME
5. Recycle time	3.3.2	4.3.4.3	SME
6. Workmanship	3.5	4.3.7	Visual

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TABLE II - CLASSIFICATION OF DEFECTS
(Use Inspection Level L5, and Sampling Plan Table V of Standard MIL-STD-105)

Major AQL 1.5	Req. Par.	Q.A. Par.	Test Method
1. Vibration	3.4.1	4.3.5	SME

TABLE III - CLASSIFICATION OF DEFECTS
(Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105)

Major AQL 1.5	Req. Par.	Test Method
1. Improper cleaning (Level A)	5.1.1.1	Visual
2. Improper drying (Level A)	5.1.1.2	Visual
3. Improper packaging method applied (Level A)	5.1.1.3	Visual
4. Improper inadequate immobilization (Level A)	5.1.1.3.1	Visual
5. Insufficient desiccant (Level A)	5.1.1.3.2	Visual
6. Humidity indicator missing (Level A)	5.1.1.3.3	Visual
7. Barrier unsealed or ruptured (Level A)	5.1.1.3.5	Visual
8. Improper inadequate cushioning material (Level A)	5.1.1.3.6	Visual
9. Non-conforming box (Level A)	5.1.1.3.6	Visual
10. Non-adherence to closure of box (Level A)	5.1.1.3.6	Visual
11. Type, class shipping container does not meet minimum requirements (Levels A&B)	5.2.1; 5.2.2	Visual & SME
12. Improper strapping (Levels A&B)	5.2.1; 5.2.2	Visual
13. Markings missing, illegible or improper (Levels A&B)	5.3	Visual
Minor AQL 4.0		
41. Humidity indicator incorrectly secured (Level A)	5.1.1.3.3	Visual
42. Improper humidity indicator (Level A)	5.1.1.3.3	Visual
43. Non-adherence to closure of box (Level A)	5.1.1.3.4	Visual
44. Improper barrier material (Level A)	5.1.1.3.5	Visual
45. Workmanship (Levels A&B)	5.5	Visual

TABLE IV - CHARACTERISTICS FOR PILOT PACK ONLY

Characteristic	Req. Par.	Q.A. Par.	Test Method
1. Heat seal	5.1.1.3	4.3.6.2	Visual
2. Quick leak	5.1.1.3	4.3.6.2	Visual

4.2.2.2 Level "C" preservation, packaging and packing.- Due to the variance of preservation, packaging, and packing that would conform to this Level of preservation, packaging and packing, the contractor must submit his intended preservation, packaging, and packing methods and inspection procedures to the procuring activity for approval and for preparation of classification of defects (CD's) and acceptable quality levels (AQL's).

4.2.3 Inspection equipment.- The test methods and procedures outlined in 4.3 shall be used as standards for determining the adequacy of the contractor's inspection equipment and methods utilized for inspecting the characteristics in Tables I,

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II and III. Unless otherwise specified in the contract, the contractor shall provide and maintain gages and other measuring and testing devices considered necessary for inspection of supplies for conformance to contracted requirements. The Government reserves the right to certify all such equipment and use this equipment for Government verification inspection.

4.3 Test methods and procedures.

4.3.1 Continuity.- A continuity check of the timer shall be made utilizing standard measuring equipment to determine conformance with the requirements specified in 3.2.1.

4.3.2 Dielectric strength.- The unit shall be tested in accordance with Method 301 of Standard MIL-STD-202 to determine conformance with the requirements specified in 3.2.2.

4.3.3 Insulation resistance.- Immediately after satisfactorily withstanding the dielectric strength test, the insulation resistance test shall be performed on the unit under test in accordance with Method 302, Test Condition B, of Standard MIL-STD-202 to determine conformance with the requirements specified in 3.2.3.

4.3.4 Performance.- To determine compliance with 3.3 the following pretest information shall apply. All symbols referred to herein are those shown on Drawing D7614285 or on Figure 1.

4.3.4.1 Switches S1 and S4 of the test circuit shall be opened. Switches S2 and S3 shall be operated to position 1. The Timer shall be connected to the test circuit.

4.3.4.1.1 Switch S1 shall be closed, and the transformer item 7 shall be adjusted for a reading of 120 volts on the voltmeter item 2. S1 shall be reopened and left open for a minimum of 30 seconds before starting tests. Timers T1 and T2 shall be set to zero.

4.3.4.2 Operate time.- On the unit under test, set the delay time potentiometer R11 to full counterclockwise position and switch S1 shall be closed. The operate timer T2 shall start at which time the relay K1 is energized, and after a period of 10 to 18 seconds, the operate timer T2 shall stop. Switch S1 shall then be reopened and operate timer T2 set to zero.

4.3.4.2.1 Set the delay time potentiometer R11 to full clockwise position and switch S1 shall be closed. The operate time as indicated by the operate timer T2 shall be within 33 ± 7 seconds, at which time the relay K1 is energized.

4.3.4.3 Recycle time.- Allow a minimum one minute between tests to permit tubes to return to a cold stable condition. The delay timer potentiometer R11 shall be adjusted so that on repeated tests, as in 4.3.4.2 and 4.3.4.2.1, the operate time as indicated by the operate timer T2 shall be within 33 ± 7 seconds.

4.3.4.3.1 With the timer interval operated, switches S1 and S2 shall be set to position 1. Operate timer T2 and interrupt timer T1 shall be set to zero. Switch S2

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shall be set to position 2 for 0.5 seconds, as indicated by interrupt timer T1. Switches S2 and S1 shall be returned to position 1. Proper operation of the Interrupt Timer T1 shall be indicated by a start and stop period. Repeating this operation, the relation between the interrupt time and the re-operate time shall fall within the limits specified in 3.3.2.

4.3.4.3.2 Switch S2 shall be set at position 1, and switches S3 and S4 at position 2. The circuit shall be interrupted by operating S2 to position 2, and then back to 1. Using the same interrupt time specified in 3.3.2, the same re-operate time limits shall apply.

4.3.5 Vibration.- The timer shall be mounted, using its own mounting provisions, on a vibrating machine capable of producing a motion at a constant frequency conforming to the frequencies specified in 3.4.1 with an amplitude of 0.030 and 0.060 inch total excursion. The testing machine shall be equipped with a medium to assure that mounting is free from resonances over the test frequency range. The medium shall hold the unit under test in the normal operating position. The unit shall be vibrated successively along each of its three mutually perpendicular directions that are respectively parallel to the edges of the unit with an acceleration, duration, and frequency conforming to the conditions in 3.4.1. The measuring instruments of the testing equipment shall be capable of measuring the vibration amplitude within ± 5%, and capable of measuring the frequency within ± 2%. The applied frequency shall be changed at a uniform rate of change in compliance with 3.4.1. Upon completion of the test, the unit under test shall be inspected to determine conformance with the requirements specified in 3.4.1.

4.3.6 Pilot pack.- The pilot pack submitted for acceptance inspection shall be inspected to determine compliance for all the requirements set forth in this specification.

4.3.6.1 Pilot pack failure.- The pilot pack shall be rejected when any of the items fail to conform to the requirements herein. Should a package fail any one of the requirements, production shall cease until the manufacturing process causing the defect has been corrected by the contractor, and the Government inspector informed of the corrective action. Reinspection shall not be permitted on previously rejected material; a new pilot pack must be selected after corrective measures have been taken. All rejected packs will be corrected by the contractor, and resubmitted as parts of the first production lot submitted for Government acceptance inspection.

4.3.6.2 Packaging test.- The inspection procedures and methods outlined in Section 4 of Specification MIL-P-116 shall apply when performing acceptance inspection for the characteristics listed in Table IV of this specification. Sampling plans, AQL values, and inspection levels specified in Specification MIL-P-116 shall be omitted.

4.3.7 Workmanship.- Workmanship shall comply with the requirements of 3.5 and 5.5.

4.4 Inspection of components and subassemblies.- Components and subassemblies shall be inspected in accordance with Table V. The acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall

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constitute a defective unit. The sampling plans listed in Table V are not intended to preclude 100% inspection by the contractor if the contractor so desires.

TABLE V - SAMPLING PLAN FOR COMPONENT AND SUBASSEMBLY INSPECTION

Lot Size (inclusive	Sample Size	Accept Lot	Reject Lot
0-25	100% Inspection	0	1
26-35	20	0	1
36-50	25	0	1
51-150	30	0	1
151-over	30	1	2

5. PREPARATION FOE DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning.- The unit shall be cleaned in accordance with Process C-1 of Specification MIL-P-116.

5.1.1.2 Drying.- Drying shall be accomplished by procedure D1 or D4 of Specification MIL-P-116.

5.1.1.3 Unit packaging.- The timer shall be packaged Method IIb of Specification MIL-P-116 and as specified herein:

5.1.1.3.1 Immobilizing in fiberboard box.- The unit shall be immobilized within a fiberboard box by slotted support blocks at either end. (See Figure 2). The fiberboard box shall conform to Type I, Class I of Specification PPP-B-636. The size of the wood support blocks shall be based upon the size of the electronic unit. The height of the container shall be sufficient to permit a minimum clearance of 1/2 inch from the top and bottom of the unit when it is positioned in the container. The length of the fiberboard box shall be such as to provide a snug fit against the support blocks when the unit is in position. Side movement shall be prevented by using fiberboard sheets between the unit and the sides of the container.

5.1.1.3.2 Desiccant.- The required units of desiccant, computed according to Formula I of Specification MIL-P-116 and conforming to Specification MIL-D-3464, shall be secured within the container.

5.1.1.3.3 Humidity indicator.- A humidity indicator, in accordance with Standard MS 20003 shall be secured within the box as far as possible from the desiccant and plainly upon opening the box.

5.1.1.3.4 Closure of fiberboard container.- The fiberboard box shall be sealed with tape conforming to Specification PPP-T-76.

5.1.1.3.5 Water-vaporproof barrier.- Corners of the box shall be blunted and the box shall be sealed within a barrier fabricated of material conforming to Specification MIL-B-131, class 1. The time lapse between securing the desiccant and the

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closure of the barrier shall be held to a minimum.

5.1.1.3.6 Cushioning.- The unit shall be cushioned within a second fiberboard box. Cushioning material shall be a minimum thickness of one inch on all surfaces and shall conform to type II of Specification MIL-C-7769. The second fiberboard box shall conform to Type II, Class I of Specification PPP-B-636. The fiberboard box shall be sealed with tape conforming to Specification PPP-T-76.

5.1.2 Level C.- Cleaning, drying, preservation, and packaging shall be in accordance with the manufacturer's commercial practice, when adequate.

5.2 Packing.

5.2.1 Level A.- A quantity of units (not exceeding 12) packaged as specified shall be packed in a shipping container conforming to either of the following specifications:

PPP-B-601, overseas type
PPP-B-621, class 2 - overseas

The container shall be strapped in accordance with appendix of the applicable box specification.

5.2.2 Level B.- A quantity of units (not exceeding 12) packaged as specified shall be packed in a shipping container conforming to either of the following specifications:

PPP-B-601, domestic type
PPP-B-621, class 1 - domestic

Requirements for strapping shall be in accordance with appendix of the applicable box specification. Staples shall not be applied to straps on domestic shipments.

5.2.3 Level C.- Packaged units shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or Motor Freight Regulations, whichever is applicable.

5.3 Marking.- Unit packages and exterior containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

5.4 Pilot pack.- Prior to quantity packaging, a pilot pack, consisting of a complete and packed shipping container shall be inspected to determine conformance with the requirements of this specification.

5.5 Workmanship.- Adequate controls shall be utilized to check for contamination of cleaning solvents. Wraps shall be neat, conforming, and firmly secured. Tape shall be uniformly applied to interior boxes and shall completely cover and adhere to all joints and seams. Shipping containers shall be free of protruding

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staples, screws, nails, etc. Strapping shall be applied perpendicular to the sides of the container just prior to shipment and shall not cover any container marking. Irrelevant marking shall be neatly obliterated.

6. NOTES

6.1 Intended use.- The purpose of the timer is to provide a time delay of approximately 22 seconds in the application of plate voltages to gas tube rectifiers after the heaters have been energized. Provisions is made to vary the time delay after brief interruptions of power, the length of delay depending upon the length of the interruption.

6.2 Application.- The timer is a component of the following assemblies and systems:

Computer 7604074	Antiaircraft Fire Control System M33C (90MM) 7620434
Computer 7603993	Antiaircraft Fire Control System M33D (120MM) 7620435
Director Station Group OA/711/MSA-7 Power Supply Group OA-742/MSA-7 Power Supply Group OA-730/MPA-4	Radar Course Directing Central AN/MSW-1
Radar Set Group 7620243	Antiaircraft Fire Control Systems M33C (90MM) 7620434 and M33D (120MM) 7620435

6.3 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of protection required.
(c) Applicable stock number.

6.4 Definitions.

6.4.1 Standard measuring equipment (SME).- Standard measuring equipment (SME) includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

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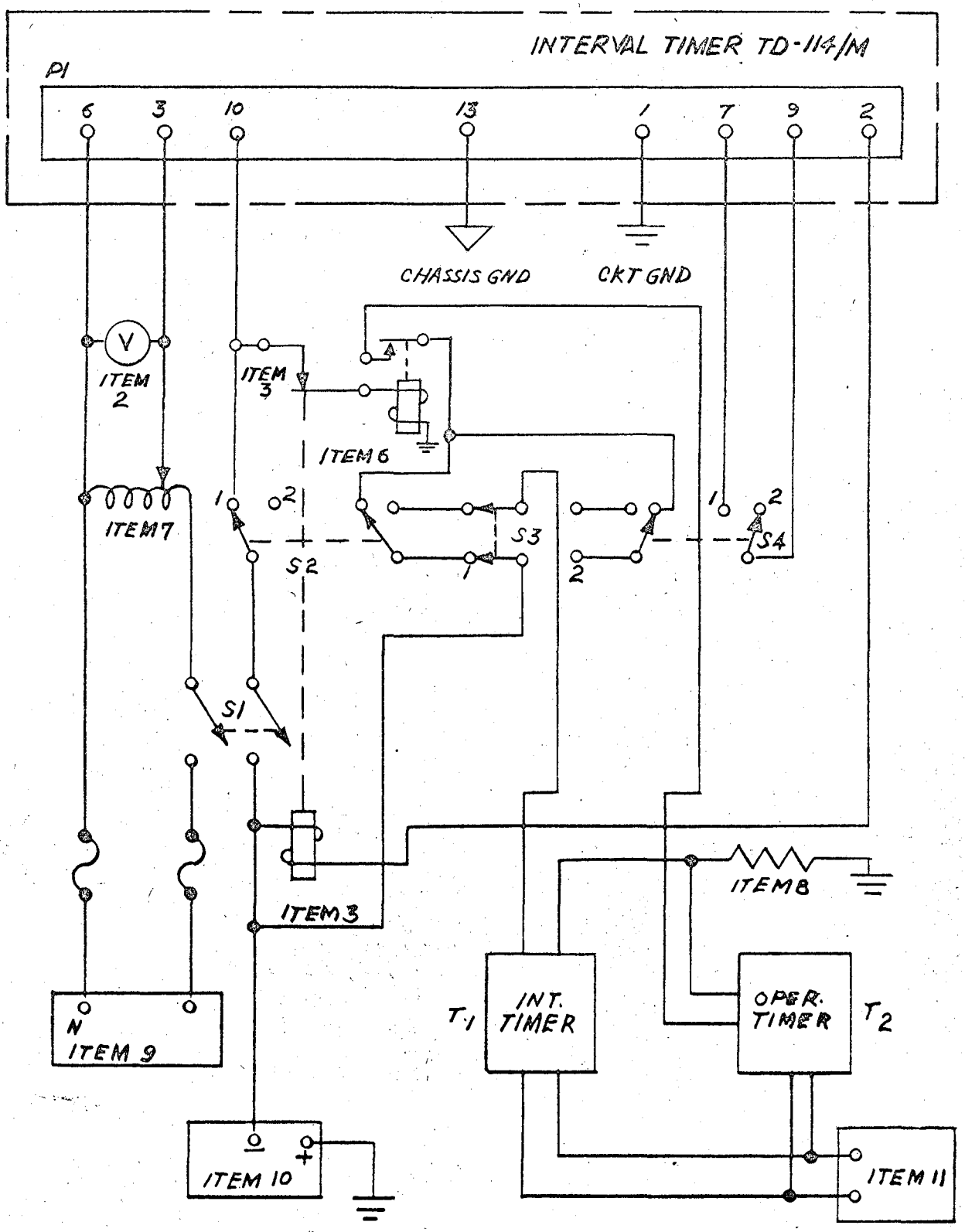
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SHEET 1 OF 2

FIGURE 1

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Item No.	Description
1.	Timers (T1, T2), solenoid operated, 0-60 seconds, 1 second sweep.
2.	Voltmeter, ac, 0-150 volts 2%
3.	Relay, 28 volts, one (1) normally closed contact.
4.	Switches (S1), double pole single throw.
5.	Switches (S2, S3, S4), double pole double throw.
6.	Relay, 28 volts, one (1) normally open contact.
7.	Transformer, 60 cycle variable voltage, 1 ampere, 120 volts.
8.	Resistor, 100 ohms 10%, 10 watts.
9.	Power supply (120 volts, 400 cycles, 1 ampere, regulated 6 volts).
10.	Power supply, (-28 volts, 1 ampere, regulated 4 volts).
11.	Power supply, (120 volts, 60 cycles, regulated 6 volts).

Figure 1 Sheet 2 of 2

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1. Electronic Unit
2. Support blocks - group I or II lumber of Spec. PPP-B-621 - 2-req'd.
3. Desiccant conforming to Spec. MIL-D-3464.
4. Humidity indicator - MS 20003.
5. Inner fiberboard box - type I, class I, style RSC of Spec. PPP-B-636.
6. Water vaporproof barrier - class I of Spec. MIL-B-131.
7. Bound cushioning material - MIL-C-7769.
8. Outer fiberboard box - W5c, style RSC of Spec. PPP-B-636.

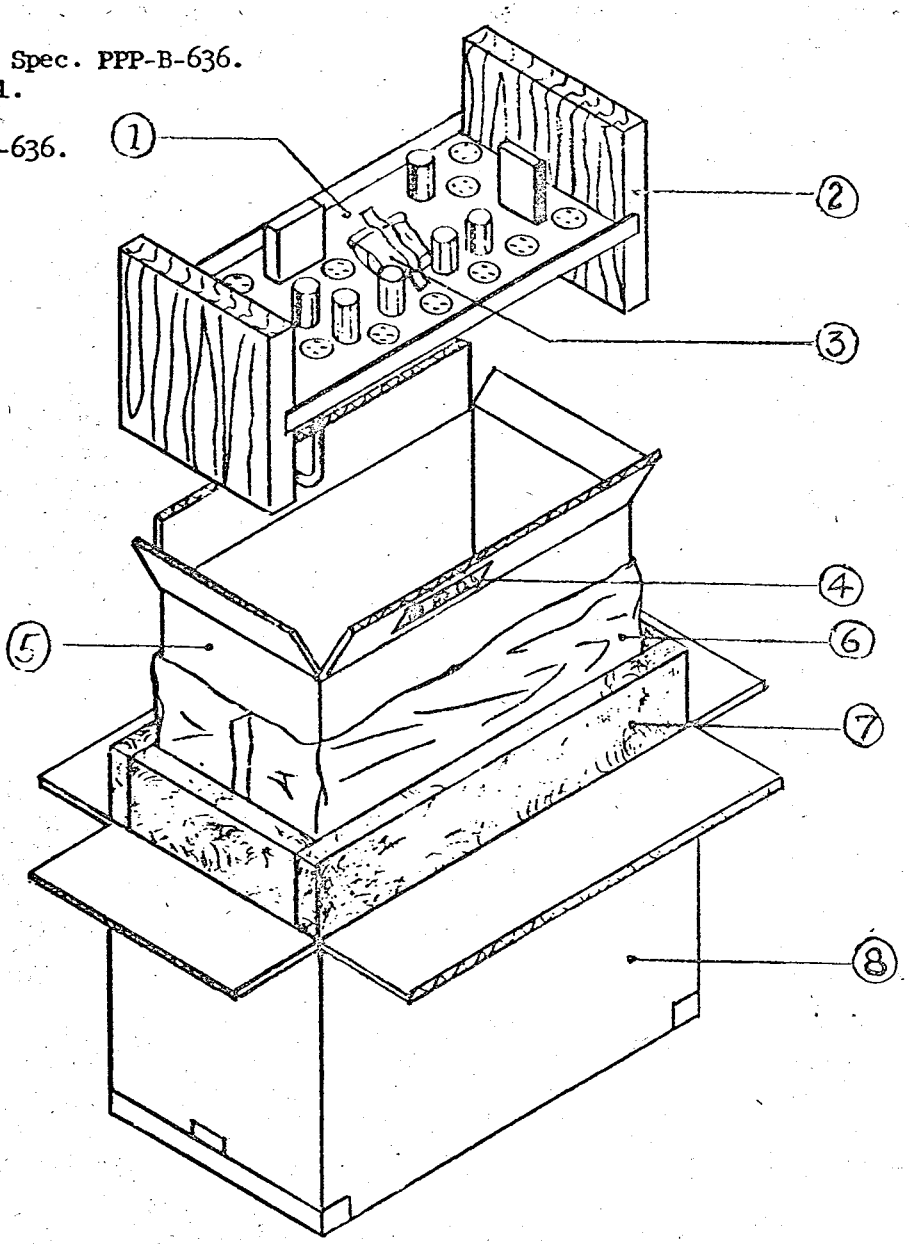


FIGURE 2

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